

Building on the active-inference driver model: what we have done at Chalmers, and what is open

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Why we started, and what this note is

Our interest in the model of Schumann et al. (2026) comes from a question it seemed well placed to answer: can a driver's comfort-zone boundary be measured with one quantity that means the same thing in every scenario? The preference prior looked like that quantity. A second interest is crash causation: whether the model can serve as the response process inside a counterfactual crash-generation model.

This note summarizes what we implemented, what we tried, what worked and what did not, and where we would value your view. The released code and the OSF deposit made all of it possible, and we are grateful for how complete they are. We have tried to state our readings as readings; you know the model far better than we do, and several of the open issues at the end are exactly the places where we may have misunderstood it.

1 The data we used

source	what it is	what we used it for
Your code, OSF	the released code, and the deposit for all three scenarios including the ablation grids	replication, and understanding the mechanisms
Video 1	clips of four scenarios — a cut-in (by a car and by a truck), a left turn across the path of an oncoming vehicle, overtaking a cyclist, overtaking a truck — frozen at fixed moments, with the question "would you intervene?"; 80 participants, 43 of whom completed all four scenarios	per-driver boundary levels, and whether they are shared across scenarios
Video 2	cut-ins only: 144 participants, 10 944 trials, 378 clips crossing speed difference with time to collision (ego 110–130 km/h, gaps 4–80 m)	the comparison of candidate criticality measures, and the gate
Test track	26 drivers making real left turns in front of an oncoming car, 2013 (Bärghman et al., 2015)	a real-driving anchor for the left-turn boundary
QUADRIIS	5 000 synthetic rear-end pre-crash scenarios with real-world weights (Wu et al., 2025); glance and deceleration distributions digitized from Bärghman et al. (2024)	crash causation

Both video studies come from a related project and are unpublished. Video 1's design ties gap and time to collision together completely, which is why video 2 was needed: it crosses them, so that clips at the same time to collision span a sixfold range of gap.

2 What we built

Replication. Your code, run on our CPU-only hardware, reproduces the deposit to the timestep; we verified the looming-threshold mechanism across all 896 deposited rear-end runs. We also wrote a readable NumPy mirror of the six preference terms, following the released code wherever it differs from the SI, and a library of surprise measures along the lines of Dinparastdjadid et al. (2023).

A practical constraint that shaped the rest. On CPU a single scenario ran from minutes to hours, with no clean predictor of which. So wherever we could, we evaluated the preference function pointwise on recorded kinematics instead of running the loop. Much of what follows depends on that shortcut, and we come back to whether it is legitimate in section 6.

3 Comfort-zone boundaries: what we tried, and what the data said

3.1 The starting idea. The *deficit* — how far the preference of the current state falls short of the preferred state — is zero inside the comfortable region and grows outside it. A boundary is then a level of the deficit, a driver's boundary is their own level, and the deliverable is a percentile of levels across drivers. It needs only the preference function and recorded kinematics.

3.2 Two pre-registered tests went against it. First, accumulating the deficit over time to time the response did not fit the repeated-clip design. We read that as a verdict on our own accumulator (deficit-driven, non-leaky, fixed bound), not on evidence accumulation in general. Second, and decisively, we asked whether the deficit is the right criticality measure at all. On video 2, with the models, folds and decision rule written down before the run, the deficit scored a held-out error of 0.347 against 0.152 for a plain gap threshold (chance 0.320; the sampling-noise floor of the data 0.118).

A later decomposition split that loss in two. About 44% came from a construction of our own — a continuous lane-entry weight we added for the cut-in (section 4), which suppressed the deficit in exactly the slow lane changes people responded to most. The rest came from the worst-case counterfactual of the safety term. In video 2's regime that counterfactual is violated almost everywhere, so the term tracks the absolute speeds: about 0.3 m/s of change per meter of gap, against 1.5–2 m/s per m/s of either vehicle's speed. Within clips of equal time to collision it therefore ranks by speed, where the raters rank by gap.

We read this narrowly, as a statement about the counterfactual *used as a comfort criterion*. For collision avoidance, where the outcome genuinely depends on where the other vehicle will be and how fast everyone is going, it may be exactly the right reference.

3.3 What worked: your model's perceptual front end. Among the candidates, the optical expansion rate $d\theta/dt = W \cdot \Delta v / (g^2 + W^2/4)$ — the quantity your perception stage computes — fitted best: 0.113, at the noise floor, against 0.152 for gap, 0.152 for optical size, 0.168 for inverse tau and 0.289 for required deceleration. A freely weighted rule in log gap and log time to collision put a weight of 0.497 on the gap, which is the looming identity rather than an approximation of it. One contrast is worth noting: Xue et al. (2018) found inverse tau the better cue for brake onsets in a driving simulator. On our video paradigm the order reverses, and whether that is the paradigm or the task is, as far as we can tell, open.

3.4 The model as it stands.

$\text{share who intervene} = \text{lapse} + (1 - \text{lapse}) \times \text{gate} \times \Phi((\text{axis} - \text{level}) / \text{spread})$

- **The gate** says whether the other vehicle counts yet: the probability that its lateral clearance, projected 3 s ahead at its current rate, falls below a minimum. The idea came from a colleague's parallel analysis. Fitted on post-onset clips only, it predicted the pre-onset clips out of sample (error 0.032). The gate is the part that changes between scenarios.
- **The axis** is scenario-specific: the expansion rate for the cut-in, the oncoming vehicle's distance for the left turn.
- **The level** belongs to the driver. On the cut-in the population median is 0.032 rad/s (about 1.8° of apparent width per second), the 80th percentile 0.066 rad/s.
- **The level looks like a trait.** Without any model, each driver's propensity to intervene correlates at +0.50 to +0.74 across the six pairs of the four scenarios, about 69% of what the data's reliability allows. The fitted levels on the cut-in and the left turn agree at a rank correlation of +0.65 over the 43 drivers seen in both.
- **Real driving.** On the test track the median comfort boundary was 2.45 s of post-encroachment time, against 2.18 s on video, and the video-fitted population predicted the drivers' real go/no-go decisions better than chance (0.200 against 0.289) with nothing refitted. Drivers were far more consistent in the car: a within-driver spread of 0.20 s against 0.86 s.

3.5 Two things tried in the last few days (preliminary).

- **Emergency braking levels against comfort judgments.** Fixing the threshold at the level where Markkula et al. (2016) found emergency braking to begin — 0.02 rad/s on the expansion rate, or 0.2 s^{-1} inverse tau — made both fits clearly worse (0.153 against 0.113; 0.224 against 0.168). The fitted 50% points sit 1.6–1.7 times higher on both variables, and Farewell's level falls near the quarter point of the judgment curve. As we read it, the comfort boundary and the onset of emergency braking lie on the same variable, with most of the comfort distribution above the emergency level; a judgment on frozen video and an executed brake are of course not the same response.

- **Surprise as the start of the response.** We asked whether surprise could mark when a situation starts to count, taken either about the other road user alone or about the whole situation including the automated ego. With a constant-velocity predictor, surprise about the other road user never registered in the cyclist overtake or the left turn, where it is the ego that moves. Whole-situation surprise did register, but only after participants had begun to respond: 0.7–1.1 s after the ego pulled out to pass, and after the clips ended in the left turn. On the cut-in, a step at the surprise onset kept every pre-onset clip closed but fitted worse than our anticipatory gate (0.145 against 0.103). As we read it, what starts a response is closer to *anticipated* conflict than to surprise. On these judgments, accumulating looming from the onset also fitted worse than a threshold on the looming rate (0.196 against 0.144).

The limitations we see ourselves: both video studies show frozen clips without self-motion, the response is a judgment rather than an executed maneuver, and the left turn is the only scenario with a real-driving check. Naturalistic data, planned with an industry partner, is meant to be the test.

4 Cut-in as a scenario

We never ran the full closed loop on a cut-in; the cost was prohibitive on our hardware. What we did was stage cut-ins in the preference function, and that alone taught us what the scenario asks of the model.

- **The other vehicle** can be a replay of recorded trajectories rather than a scripted maneuver, which avoids anything like the oncoming scenario's incursion optimizer.
- **The desired speed** has to come from the clip. At the default of 15 m/s against an ego speed of 30.5 m/s, the speed term dominated everything else.
- **The lateral tests are binary.** Looming is perceived within 3 vehicle widths, and the collision, safety and closing-rate terms apply within 1.15 widths. A cutting-in vehicle straddles the boundary for about 2.5 s, and none of the three published scenarios sustains that state. Since the safety term is also an indicator, two clips of very different urgency (a 10 m and a 21 m gap at 30.5 m/s) received an identical safety score, switching on as a step when the box test flipped.
- **Our fix, and what it cost.** We made lane entry continuous by projecting the lateral overlap to the moment of longitudinal closure. That is the construction behind the 44% share of the loss in section 3.2. Projecting over a fixed 3 s horizon instead removed that share.
- **The norms do not fit, and a proposal.** None of the three published norm sets suits a vehicle that is *transiently and legitimately* straddling the line. A positional norm withdraws trust as the body leaves its lane, and on our 90 recorded lane changes that moment comes later for slower ones (a median 0.53, 0.73 and 0.93 s after the change registers, for 2, 3 and 4 s lane changes), which is the pace dependence participants do not show. We would propose a *crossing norm*: either lane is normal, and straddling is normal while crossing toward the new lane at a plausible lateral speed, but not when stalling, drifting back or swerving. It needs only the state the particles already carry. A one-page proposal is attached; nothing has been run in the loop.

5 Crash causation

We placed your model as the response process inside the crash-causation model of Bärghman et al. (2024), keeping its causation components as switchable parts: off-road glances placed as a renewal process, too-close following, a deceleration cap, a no-response share, and the abnormal-acceleration follower of Wu et al. (2025). Crashes were generated from the QUADRIS seeds, with the follower's original speed profile minus its own braking as the counterfactual, and compared with the QUADRIS reference by the practical-equivalence method of Wu et al. (2026).

- **Severity.** The active-inference conditions came closer to the reference crash severities than the original model's fixed-delay response (θ 0.148 against 0.209), despite producing 4.7 times the crash probability. Tested as a difference the ordering holds with a probability of about 0.97. Neither reaches practical equivalence.
- **Braking.** Only the active-inference conditions reproduce the reference's many crashes with no follower braking; the weighted mean follower braking came within 0.3% of the reference, where the fixed-delay response over-braked by 35%.
- **Timing.** Brake onsets came a median 1.25 s after the anchor (interquartile range 0.50–1.75 s), against the fixed 0.50 s. A cheap open-loop surrogate — the pointwise preference plus the accumulator — matched the full closed loop to a median of 0.55 s across 23 scenarios from 1.3 to 35.5 m/s.
- **The glance gate.** Forcing off-road glances through the code's own observation gate, a driver who had registered the lead's braking kept responding *during* the glance, close to the attentive onset, even under a near-total blackout. As we understand it, the belief cloud coasts on its own norm-shaped prediction and the accumulator

keeps filling. The fixed-delay model assumes the opposite, so the two differ testably when a glance begins after the conflict has been registered.

- **A caution about validation.** In this crash population severity and timing are close to independent: roughly 70% of the variation in impact speed comes from the scenario rather than the response. A model validated only on severity may therefore be close to unconstrained in its response timing.
- **Letting the model choose its glances (preliminary).** We asked whether the planner's "avoid off gaze" override could be lifted and the gaze parameters fitted to the SHRP2 glance distribution. Lifting it is a one-line change, but the probe ran into something else first: in steady following with nothing happening, the released configuration began braking after 3.2 s at a 1.5 s headway and 4.6 s at 2.0 s, and in most repeats braked to a standstill — with gaze choice on or off, and with your own scripted lead. With perception noise a hundred times larger, it followed steadily and never looked away. Glances, where they occurred, always lasted one 0.2 s step, against a SHRP2 median of about 0.75 s, and a near-blind glance was chosen about as readily as an ordinary one. As we read it, fitting glances would first need a reason to look away, a cost that depends on perception noise, and a way for a glance to last; we have not tried any of those.
- **Why it brakes (preliminary).** Reading the planner's own bookkeeping, the braking is a single evidence-triggered re-plan at about 2.6 s. The evidence comes entirely from imagined collisions and failed safety checks in the followed plan — 68 090 per step, which matches the value for this condition in your OSF deposit — and the new plan trades speed, pedal effort and lane position for safety margin. The inverse-tau preference and the epistemic term play no part. Raising the noise on the state channels alone slows the accumulation 30-fold and the car keeps following; raising it on the looming channels alone does not. The published runs never show this, since the lead brakes 0.6 s in.

6 The questions we would most like to ask

- 1 **Comfort or collision?** Was the safety-margin counterfactual meant to describe what drivers find comfortable, or only what avoids a collision? If comfort is in scope, would you model it as a level of the same preference, or against a different reference?
- 2 **The closing-rate preference.** The one-sided inverse-tau term takes its center of 0.2 s^{-1} from Markkula et al. (2016), where drivers in naturalistic rear-end emergencies seldom braked before looming reached that level and mostly braked within a second after. In the model it becomes the center of a *preference*, which, as we read it, makes it close to a comfort standard. Was that intended, and where does the spread of 0.125 s^{-1} come from? Our comfort judgments put the 50% point at 0.33 s^{-1} on the same variable (section 3.5).
- 3 **Perception noise, and where the glance multiplier acts.** Above the looming threshold the observation noise is $0.001 \times 0.01 = 10^{-5}$, which makes perception essentially exact once the threshold is crossed: closing at 5 m/s, the threshold is crossed at 63 m and the distance is then known to about $\pm 2 \text{ cm}$. Was the factor 0.01 deliberate? The off-road multiplier of 3 is applied to that noise *before* the threshold override, so below the threshold a glance changes nothing and above it 10^{-5} becomes 3×10^{-5} . As we read it, a glance then costs almost no information. Was that the intended order?
- 4 **A graded perception switch.** Both the looming detection threshold and the lateral applicability tests (3 widths for perception, 1.15 for the preferences) are on/off. Did you consider graded versions? Our own inclination would be a psychometric detection function at the detection level, and to keep a Farewell-like transition at 0.2 s^{-1} out of perception, since that level describes responding rather than seeing — perhaps as a gain on accumulation instead.
- 5 **Individual differences.** We find one per-driver comfort level shared across scenarios. Which parameter would you make per-driver: the reaction-time budget, the assumed worst-case braking, the accumulation rate, or a preference center such as the inverse-tau mean?
- 6 **A norm for a vehicle changing into our lane.** How would you write it? Our crossing-norm proposal (attached) keys straddling on lateral speed. Would you rather use a clock, or inferred intention?
- 7 **Graded lateral entry.** Did you consider weighting the collision and safety terms by lateral overlap instead of the 1.15-width box?
- 8 **The onset from inside the model.** When the norm tournament's "now" weight collapses, the model effectively stops trusting the other vehicle, and its four-second held projection already looks ahead. Could that projection play the role of our anticipatory gate, and put the onset inside the model?
- 9 **Glances, and sustained following.** Is the coasting of beliefs through a glance intended? And would the gaze machinery of Engström et al. (2024) be ready to switch on here? In a quick probe (section 5) we had two surprises:

the released configuration did not hold sustained car following — it began braking after 3–5 s with nothing happening — and glances, once allowed, always lasted a single time step, since looking back is instantaneous. Is sustained following something you have looked at? The braking traces to the accumulator re-planning on imagined collisions, and noisier perception of the car's own state prevents it — was the small perception noise chosen with that in view? And how would you give the model a reason to look away, and a reason to stay away?

- ¹⁰ **Practicalities.** Is the pointwise evaluation of section 2 a fair approximation in your view, is there a faster implementation we should use, and would you be interested in a joint look at cut-in data? For naturalistic data, where nothing marks the start of an event, what starting point for the accumulator would you use?

References

- Bärgman, J., Smith, K., & Werneke, J. (2015). Quantifying drivers' comfort-zone and dread-zone boundaries in LTAP/OD scenarios. *Transportation Research Part F: Traffic Psychology and Behaviour*, 35, 170–184.
- Bärgman, J., Svärd, [initials], Lundell, [initials], & Hartelius, [initials] (2024). [Title and source to be completed.]
- Dinparastdjadid, A., Supeene, I., & Engström, J. (2023). *Measuring surprise in the wild* (arXiv:2305.07733). arXiv. <https://arxiv.org/abs/2305.07733>
- Engström, J., Wei, R., McDonald, A. D., Garcia, A., O'Kelly, M., & Johnson, L. (2024). Resolving uncertainty on the fly: Modeling adaptive driving behavior as active inference. *Frontiers in Neurorobotics*, 18, Article 1341750. <https://doi.org/10.3389/fnbot.2024.1341750>
- Markkula, G., Engström, J., Lodin, J., Bärgman, J., & Victor, T. (2016). A farewell to brake reaction times? Kinematics-dependent brake response in naturalistic rear-end emergencies. *Accident Analysis and Prevention*, 95, 209–226. <https://doi.org/10.1016/j.aap.2016.07.007>
- Schumann, J. F., Engström, J., Johnson, A., O'Kelly, M., Messias, J., Kober, J., & Zgonnikov, A. (2026). Active inference as a model of collision avoidance behavior in human drivers. *Nature Communications*, 17, Article 5009. <https://doi.org/10.1038/s41467-026-73345-0>
- Wu, [initials], Flannagan, [initials], Sander, [initials], & Bärgman, J. (2025). [Title to be completed.] *IEEE Transactions on Intelligent Transportation Systems*. arXiv:2406.15538.
- Wu, [initials], Sander, [initials], Flannagan, [initials], & Bärgman, J. (2026). *Practical validation of synthetic pre-crash scenarios* [Preprint; full reference to be completed].
- Xue, Q., Markkula, G., Yan, X., & Merat, N. (2018). Using perceptual cues for brake response to a lead vehicle: Comparing threshold and accumulator models of visual looming. *Accident Analysis and Prevention*, 118, 114–124.